GeoCalib: Supplementary Material

Anonymous submission

Untouched Dataset/Source Transfer Stress Test

We prospectively froze an untouched dataset/source target before source prediction: the 11-scene official ReplicaSSG test split and the official VisualGenome-trained FROSS RT-DETR-EGTR release. ReplicaSSG validation scenes were excluded from method, mapping, threshold, and score selection. The exact mapping is `near`→`close by`, `above`→`higher than`, and `under`→`lower than`; support/contact predicates are excluded because no exact subtype mapping was available. The run contains 4,290 candidates, 172 exact-label GT relations, 11 paired bootstrap units, and passes all 24 frozen provenance/firewall checks.

Table 1: **Negative ReplicaSSG/FROSS transfer at $K = 100$.** Deltas and 95% CIs are paired over the same 1,000 scene-bootstrap samples. The frozen joint gate requires the recall CI lower bound to exceed -0.01 and the violation CI upper bound to be below zero.

Condition	R	ΔR [95% CI]	V	ΔV [95% CI]
Semantic	.3605	–	.1967	–
Product	.3605	.0000 [.0000,.0000]	.1967	.0000 [.0000,.0000]
Rank-average	.3314	$-.0291$ [$-.0741$,.0133]	.0384	$-.1583$ [$-.1929$,$-.1219$]
RRF ($c = 60$)	.3314	$-.0291$ [$-.0579$,$-.0051$]	.0595	$-.1372$ [$-.1682$,$-.1045$]

Neither frozen GeoCalib instantiation passes. Product is identical to semantic ranking at the primary endpoint, so it fails the strict violation-improvement gate. Rank-average sharply lowers verifier-derived V but fails the recall guardrail. Product improves secondary $K = 20$ and $K = 50$ tradeoffs, but these do not replace the preregistered $K = 100$ decision. We therefore retain SGFN as positive source-level prospective evidence and treat ReplicaSSG/FROSS as a transparent negative-transfer result and limitation, not as dataset-level confirmation. V remains verifier-derived rather than independent human physical validity.